

TIDE MACHINE — RAILS AND FRAME drilling templates, 1:1

1×2 pine, 19.05 × 38.1 actual. Two rails 470 mm long plus two 140 mm end posts.
Templates, not a cut file — tape a sheet to the wood, centre-punch every cross, then drill.
Sheets 2–4 carry the rail at 1:1 in three tiles with a 30 mm overlap; each tile shows all three faces.
Print at 100%, trim each tile on its red registration line, and tape the run together before marking anything.

THE TWO RAILS ARE MIRROR IMAGES. Use the template with the matching name.

Both rails carry the same features at the same machine (x, z) — but each is drilled on its INNER face, the one that looks at the other rail. Laying the top rail inner-face-up turns it over, so with the rear edge toward you on both, machine x runs left-to-right on one rail and right-to-left on the other. Drill both from one template and the second rail is scrap. Each template names its rail and states which way x runs. Register the REAR EDGE first, then the end.

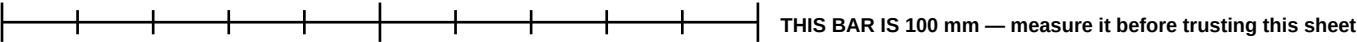
Drill schedule

- | | |
|--------------------------|---|
| 7 × Ø5.2 × 9 deep, BLIND | guide-rod sockets, 20 mm from the rear edge |
| 12 × Ø2.0 × 9 deep | screw-eye pilots, 30 mm from the rear edge |
| 1 × Ø7.0 THROUGH (19.05) | drum shaft at x +249.5, both rails |
| 5 × Ø2.0 × ~10 deep | faceplate pilots — REAR face, 6 mm in from the inner edge |

The screw-eye pilots leave only 8.1 mm of wood to the front edge — drill them square or pine will blow out.

Cut list

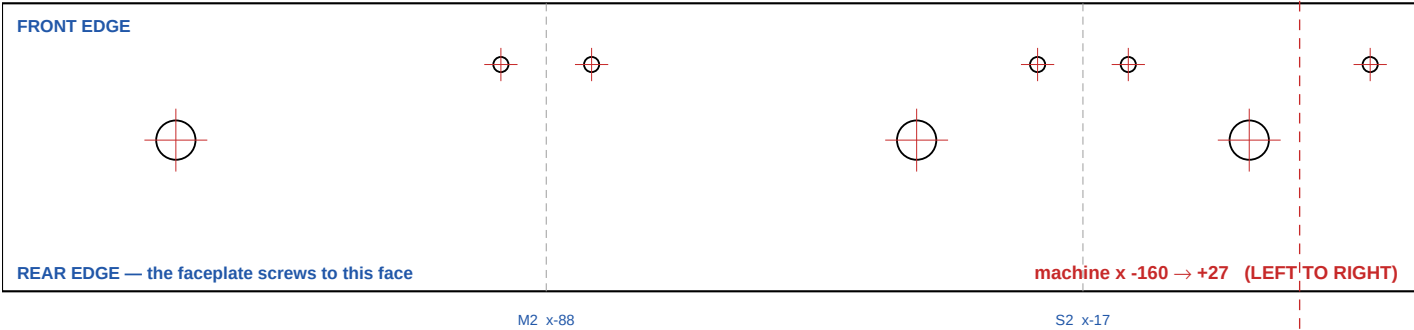
- 2 × rail 1×2 pine, 470 mm (19.05 × 38.1 actual)
2 × post 1×2 pine, 140 mm — stands between the rails
7 × rod Ø5 mm, 158 mm — socket bottom to socket bottom
One 8 ft 1×2 covers all four wood parts (1220 mm needed).



RAILS 1:1 TILE 1 of 3 machine x -160 to +27 print at 100%, do not scale

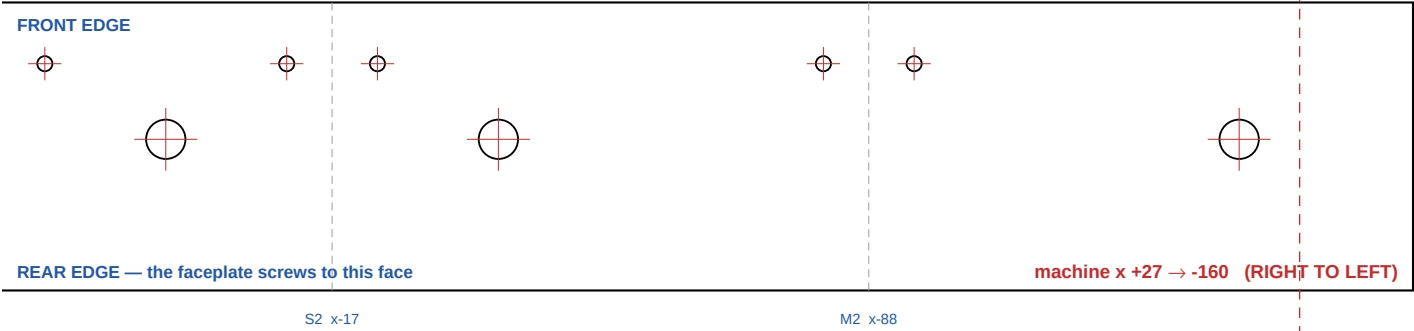
TOP RAIL — inner face

inner face, 470 × 38.10. 7 blind rod sockets, 12 screw-eye pilots, 1 through hole for the drum shaft.



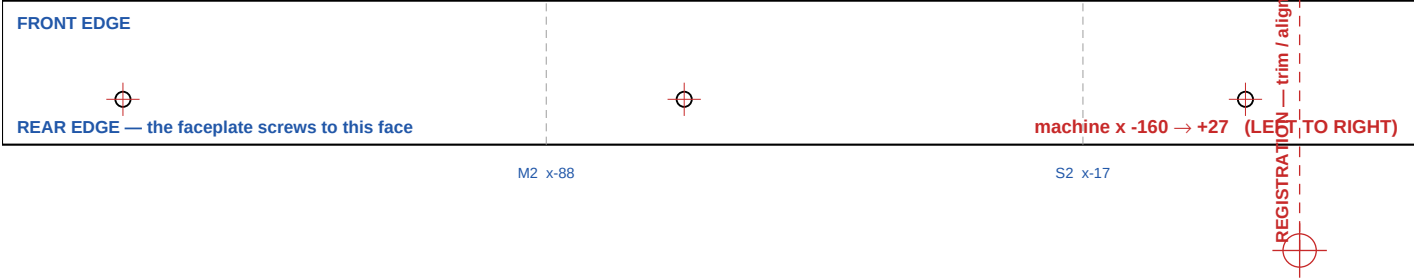
BOTTOM RAIL — inner face

inner face, 470 × 38.10. 7 blind rod sockets, 12 screw-eye pilots, 1 through hole for the drum shaft.



EITHER RAIL — rear face

rear face, 470 × 19.05. The SAME for both rails — the pilots sit 6 mm in from the inner edge either way.

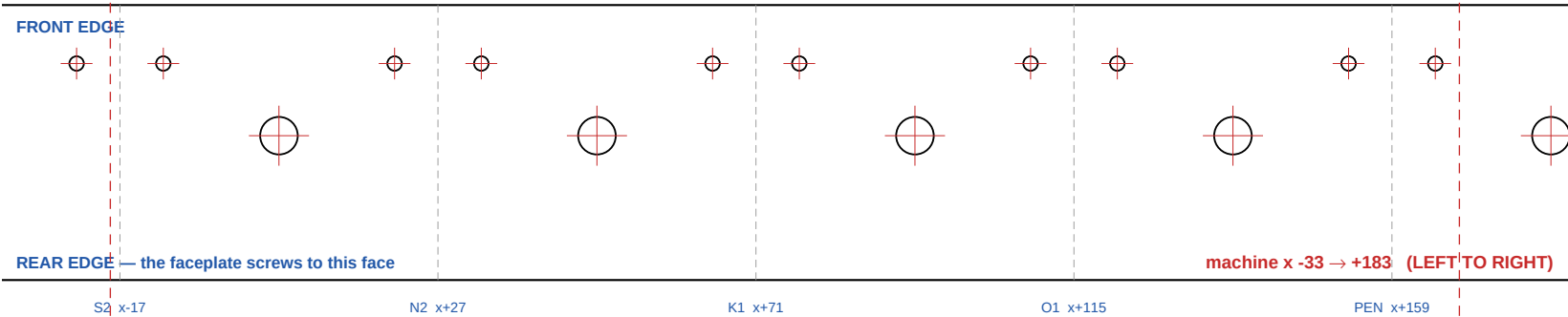


THIS BAR IS 100 mm — measure it before trusting this sheet

RAILS 1:1 TILE 2 of 3 machine x -33 to +183 print at 100%, do not scale

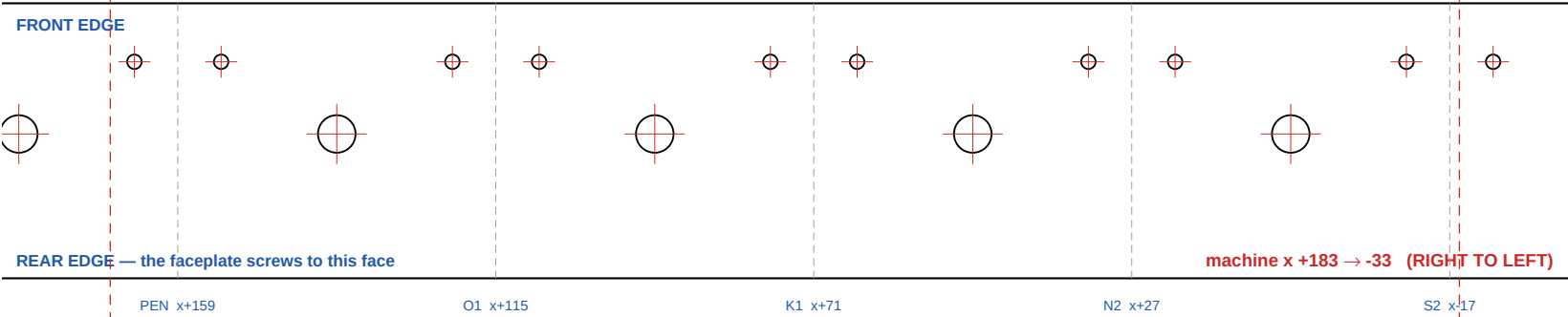
TOP RAIL — inner face

inner face, 470 × 38.10. 7 blind rod sockets, 12 screw-eye pilots, 1 through hole for the drum shaft.



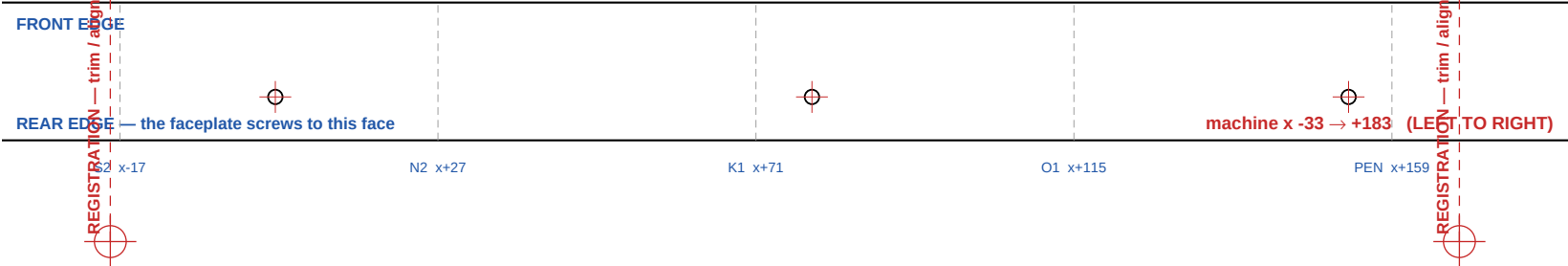
BOTTOM RAIL — inner face

inner face, 470 × 38.10. 7 blind rod sockets, 12 screw-eye pilots, 1 through hole for the drum shaft.



EITHER RAIL — rear face

rear face, 470 × 19.05. The SAME for both rails — the pilots sit 6 mm in from the inner edge either way.



THIS BAR IS 100 mm — measure it before trusting this sheet

RAILS 1:1 TILE 3 of 3 machine x +123 to +310 print at 100%, do not scale

TOP RAIL — inner face

inner face, 470 × 38.10. 7 blind rod sockets, 12 screw-eye pilots, 1 through hole for the drum shaft.



BOTTOM RAIL — inner face

inner face, 470 × 38.10. 7 blind rod sockets, 12 screw-eye pilots, 1 through hole for the drum shaft.



EITHER RAIL — rear face

rear face, 470 × 19.05. The SAME for both rails — the pilots sit 6 mm in from the inner edge either way.



THIS BAR IS 100 mm — measure it before trusting this sheet